AERIS Expert Labeling Guide

This guide explains how to fill out a blinded AERIS claim packet. The example packet that comes with it is only for checking the format. Nothing counts as an official label until my freeze checks pass.

What your labels are for

Every claim in this packet was written by a large language model. Each model was shown one anomaly and the stored measurements around it, then asked to reason through the physical cause in four steps. The claims are the sentences that came out. They are what I am testing, not conclusions AERIS stands behind.

AERIS runs two automatic checks on every one of those sentences. The first asks whether the sentence is actually grounded in the data the model was shown. The second scores each claim against independent measurement channels using fixed physical rules, with no language model and no person involved.

That second score is meant to work without an expert in the loop. Before I can claim it does, I have to test it against real expert judgment once. Your marked packet is that test. Your labels become the testing dataset, and the number I report is how often the automatic score agrees with you. The scoring pipeline never sees your labels, and none of its rules or thresholds are tuned on them, so the comparison stays honest.

This is a one-time validation, not a review step that lives on in the product. The finished product runs by itself. It spots an unusual reading, has a model explain it, checks the explanation automatically, and shows the user the explanation next to its verification result.

What's in the packet

Each case has four parts, in this order:

1. **Anomaly.** The basics of the event: data source and metric, the observed value against the expected baseline in the metric's own unit, the method that baseline came from, UTC time, location, severity, and which detectors flagged it. The baseline is the detector's own reference level, usually the mean of the preceding seven days of that series, so for a metric the network reports near or below its detection limit it can be negative.
2. **Stored evidence.** Three tables, then the station map and the wind panels, all drawn from the same stored data.
3. **Claims.** The statements you are labeling, each with its own row of V / I / U boxes.
4. **Optional: most likely cause.** A blank box for your own best guess at what caused the event. You can skip it.

The three evidence tables are look-up tables, not required reading. You never have to read them through. They are there for the moment a claim quotes a number and you want to check it. All three list one row per source and metric, so you can check a figure by finding the row it came from.

- **Stored evidence summary** gives each metric's low and high values over the stored 72 hours, its mean, and the single reading closest to the event, with how far that reading sits before or after the event.

- **Six-hour averages** gives the same metric averaged in 6-hour blocks across the window, so you can see how it moved rather than only where it ended up.
- **Per-site averages** gives the metric averaged at each site separately, with that site's distance from the event in brackets.

The last two are there because they are exactly what the models were given. A claim that quotes a 6-hour average or a distance is something you can check rather than have to take on faith.

If two models wrote the same claim word for word, it is printed once. The claims are shuffled, and the shuffle is different for each labeler, so your packet and mine are in different orders. Each claim has a number printed next to it, and that number is how I will refer to the claim if I ever need to ask you about one.

You will never see which model wrote a claim, how confident that model was, or any score from my pipeline. That is deliberate. Nothing in the packet should nudge you toward or away from what the software already thinks.

The data sources, briefly

These names show up in the tables and on the plots:

- OpenAQ: a worldwide aggregator of government air quality monitors. I only include stations I could verify as real regulatory monitors.
- TCEQ: Texas state monitoring stations (NO_2, SO_2, CO), read from the commission's public site.
- PurpleAir: cheap optical PM2.5 sensors that people mount on their houses. Dense coverage, noisier data. They go through their own quality screen.
- ASOS: the automated weather stations at airports. Direct anemometer and thermometer readings.
- OpenWeather: a commercial weather product that blends several inputs, including station data.
- NOAA GFS: the global forecast model. I use its f000 analysis fields (winds, boundary layer height). It assimilates surface observations, so it is not independent of ASOS.
- Sentinel-5P: the ESA satellite that measures column NO_2, SO_2, and other gases from orbit.
- EPA AQS: EPA's official archive of ground chemistry (NO_2, SO_2, CO). Used only for the historical window, not live collection.

Label definitions

These definitions are reprinted at the top of every packet's claims section, word for word, so you do not have to keep this guide open while you mark.

Mark each claim you answer with exactly one of these:

- **Valid (V):** the claim holds up at the precision it states, and the evidence in the packet supports it.
- **Invalid (I):** the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
- **Unsure (U):** you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Two rules follow from those:

- Missing or thin evidence means Unsure, not Invalid.
- A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

How to mark claims

You can annotate the PDF digitally, or print it, mark it by hand, and scan it back. Either works, and both count the same.

The V / I / U boxes and the note lines are real PDF form fields, so you can click a box and type in a note rather than drawing marks with a markup tool. They are plain checkboxes, not a linked set: ticking V does not clear I, so if you change your mind, untick the first box. That is deliberate, because leaving all three blank has to stay available as the way to skip a claim. The boxes and lines are also printed into the page itself, so if your reader does not offer form fields you can still mark the packet exactly as before, and nothing is lost.

One thing worth doing once: after you save, close the file and reopen it to check your marks are still there. A few PDF readers show typed marks without writing them into the file, and that check catches it while it is still fixable.

1. Mark exactly one box (V, I, or U) for each claim you answer.
2. To skip a claim, leave all three of its boxes blank. I record a blank as missing. I never quietly turn it into Unsure.
3. For Invalid or Unsure, add a short note when the reason is not obvious. Notes on Valid are welcome but optional.
4. Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict. An all-correct compound claim is Valid, and one whose parts are all contradicted is Invalid. Mark Unsure when the parts would earn **different** verdicts, when the wording is ambiguous, or when the claim leans on something it never actually states. Add a few words saying which part drove your call. Every claim stays in the study, even the messy ones.
5. The "most likely cause" box at the end is optional, and "insufficient evidence to determine a single cause" is a perfectly fine answer.

Reading the plots

The station map: the black star is the anomaly, and the colored dots are every station that contributed data in the stored 72-hour window. The legend gives the station count per source. The dashed rings are great-circle distance from the event, labeled in km at the top of each ring, so you can read a station's distance straight off the map. A ring that would fall outside the plotted area is left off. The rings look slightly oval because the axes are longitude and latitude, and at this latitude a degree of longitude is shorter than a degree of latitude. They are still true constant-distance curves. There is no basemap on purpose. Nothing on this map is context the models did not have.

The wind panels: one panel per wind source (ASOS, GFS, OpenWeather). Each arrow is one station's wind reading closest to the event time, drawn pointing downwind. Longer arrows mean faster wind, but each panel has its own scale, so compare arrow lengths only within a panel, never across them. The label above each panel gives the speed of its longest arrow. ASOS and OpenWeather arrows come from stored speed and direction values. GFS arrows come from its stored u and v wind components.

Everything in the three tables and on both plots comes from the same stored data the models saw. An automated audit re-derives all of it from the database and refuses to release a packet that has drifted, including one that has dropped any of the 6-hour or per-site rows.

Calm-wind flags

Some wind-direction claims have an orange flag printed under them. The flag means at least one relevant wind source reported an event-time speed below that source's calm cutoff, so there may not have been a stable wind direction to reason from. (Transport claims check GFS, OpenWeather, and ASOS. Point-source claims check GFS and OpenWeather.) The flag shows each affected source's event-time speed, the cutoff it used, and how many wind readings that cutoff came from, so you never have to compute anything.

The flag is information, not a verdict. A flagged claim is still yours to judge, and you can mark it V, I, or U like any other.

The cutoff is worked out separately for each wind source. Take that source's wind speeds over the stored 72-hour window and compute mean - 2*SD (population SD). If that comes out below 1.5 m/s, use 1.5 m/s instead. So the cutoff is max(mean - 2*SD, 1.5 m/s), and a claim is flagged only when the event-time speed is strictly below it. Dr. Annalisa Bracco confirmed the 1.5 m/s floor on 24 July 2026, so this rule is final.

Missing or stale wind data never creates a calm flag. It falls under the normal thin-evidence-means-Unsure rule instead.

Returning the packet

The marked-up PDF you send back is the official record, and I keep the file exactly as you return it. I type your marks into the database twice, on two separate days, without looking at the first pass while doing the second, then compare the two runs and settle any mismatch against your PDF. If one of your marks is ambiguous I will not guess what you meant. I will email you a short numbered list of the unclear ones and wait for your answer. Claims you skipped stay recorded as missing, exactly as you marked them.

If anything changes

This guide is frozen once official labeling starts. If any definition, threshold, or step in how I record your marks has to change after that, everything stops. I tell you exactly what changed and rerun the earlier labels through the pre-declared sensitivity check to show whether the change would have flipped any of them.

When labeling starts

Every rule here is settled. Official labeling starts once my remaining freeze checks pass and I send you the official packet. Until then, nothing you mark on the example packet counts as a label.